Question		Answer	Marks	AO	Guidance
16	(a)	**DR** $\int \mathrm{d}P = \int (72t - 108)\mathrm{e}^{-0.8t}\,\mathrm{d}t$ **oe**	**B1**	**3.1a**	separation of variables; condone omission of integral signs or of dt ; allow P instead of $\int \mathrm{d}P$ on left hand side
		$\frac{-108}{-0.8}\mathrm{e}^{-0.8t}$ **oe seen**	**B1**	**2.1**	
		$72\left[t\frac{\mathrm{e}^{-0.8t}}{-0.8} - \int \frac{\mathrm{e}^{-0.8t}}{-0.8}\mathrm{d}t\right]$ **oe**	**M1***	**1.1**	integration by parts; allow sign errors only
			A1	**1.1**	all correct;
		$[P =] - 90t\mathrm{e}^{-0.8t} - 112.5\mathrm{e}^{-0.8t} + 135\mathrm{e}^{-0.8t}[+c]$ **oe**	**A1**	**1.1**	allow omission of + c; allow unsimplified
		$58.5 = -90 \times 0 \times \mathrm{e}^{0} + 22.5\mathrm{e}^{0} + c$ **oe**	**M1dep***	**3.3**	may be awarded after rearrangement; must see constant of integration; **NB** if expression incorrect, need to see full substitution for **M1**
		$P = 36 + 22.5\mathrm{e}^{-0.8t} - 90t\mathrm{e}^{-0.8t}$	**A1**	**1.1**	allow just $c = 36$ as long as "$P =$" seen following completion of integration; allow eg $P = 36 + 45(\frac{1}{2}e^{-0.8t} - 2te^{-0.8t})$; like terms must be collected and fractions must be in lowest terms
		Alternative method $\int \mathrm{d}P = \int (72t - 108)\mathrm{e}^{-0.8t}\,\mathrm{d}t$ **oe**	**B1**		separation of variables; condone omission of integral signs or of dt ; allow P instead of $\int \mathrm{d}P$ on left hand side
		$\frac{\mathrm{e}^{-0.8t}}{-0.8}$ **oe seen**	**B1**		may be embedded in integration by parts

Question		Answer	Marks	AO	Guidance
		$(72t-108)\frac{e^{-0.8t}}{-0.8}-\int 72\frac{e^{-0.8t}}{-0.8}dt$	**M1***		integration by parts; allow sign errors only
			A1		all correct
		$[P=]-90te^{-0.8t}-112.5e^{-0.8t}+135e^{-0.8t}+c$	**A1**		all correct; allow omission of + *c*; allow unsimplified
		$58.5=-90\times 0\times e^{0}+22.5e^{0}+c$ **oe**	**M1dep***		may be awarded after rearrangement; must see constant of integration
		$P=36+22.5e^{-0.8t}-90te^{-0.8t}$	**A1**		allow just $c=36$ as long as "$P=$" seen following completion of integration; allow eg $P=36+45(\frac{1}{2}e^{-0.8t}-2te^{-0.8t})$; like terms must be collected and fractions must be in lowest terms
			[7]		
16	**(b)**	**DR**			
		as $t\to\infty$, $(k)e^{-0.8t}\to 0$	**M1**	**3.4**	or $\lim_{t\to\infty}(k)\,e^{-0.8t}=0$; *k* is a constant; allow eg as *t* tends to infinity / as *t* becomes very large / in the long term, $(k)e^{-0.8t}$ tends to / converges to / approaches zero **oe** do not allow eg substitution of $t=\infty$
		so $P\to 36$. [Hence the model predicts that the population of puffins] does **not** recover to 58.5 thousand	**A1**	**1.1**	condone *P* = 36
			[2]		